

A tuning-free heaped data density estimator with an identifiability limit

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ABSTRACT

Heaped data, whose recorded values concentrate on round numbers, are pervasive in survey microdata, instrument readouts and quantized machine-learning inputs, and they defeat kernel density estimation by rendering the rounding marks as spurious modes. Rounding to a grid of width D is box convolution followed by lattice sampling, so the rounded variable's characteristic function is an aliased sum of box-tapered replicas. Dividing out the box factor, a continuous generalization of Sheppard's correction, recovers that function inside a grid-Nyquist band up to an aliasing bias controlled by the mass beyond it. Past the band the density is not determined by the data, a limit stated and proved here. From this follow a de-heaping estimator with no user-set bandwidth, a superposition variant carrying a smooth base across it, and a combined estimator selecting between them automatically. The grid width, heaped fraction and grain mixture are read from the comb, each within a regime the limit sets, mapped against known truth. Against dedicated heaped-data and measurement-error methods the combined estimator is competitive or better in most regimes, and is beaten at the coarsest multimodal grids, where the data alone do not determine the density, and at the finest, where the correction adds variance.

Introduction

Recorded data are often a coarsened version of the quantity of interest. Survey respondents round ages, incomes and counts to convenient figures, instruments report to a fixed resolution, and retrospective reports concentrate on multiples of five, ten and twenty. This concentration of probability mass on a lattice of round values, called heaping, is pervasive in microdata.

This matters beyond survey statistics, because density estimation is a foundational primitive far outside the field in which heaping was first studied. Kernel density estimation and its relatives underpin generative modeling, anomaly detection, density-based clustering and simulation-based inference^{1,2}, and their inputs are increasingly quantized rather than continuous. Low-precision sensors report on a coarse grid, quantization to low bit-width is standard for efficient neural-network inference³, differentially private mechanisms discretize outputs to bound disclosure⁴, and administrative and health-record data carry the digit-preference heaping documented for self-reports^{5,6}. In each case a density model is fit to values snapped to a grid, and an estimator that takes them at face value inherits a comb of spurious modes that propagates downstream. An estimator robust to heaping, which also says when de-heaping is and is not possible, is therefore of broad practical value.

Kernel density estimation has no defense against heaping. Its accuracy is governed almost entirely by one bandwidth, and on heaped data that bandwidth faces a difficult choice: narrow enough to resolve real structure, it reproduces the comb as spurious modes at the round numbers, and wide enough to erase the comb, it also erases genuine structure near the rounding scale. Adaptive and plug-in selectors do not help, because to them the heaping looks like sharp real structure a good bandwidth should preserve.

The heaped-data literature splits into three lineages: classical grouping corrections beginning with Sheppard's, which adjust the moments of grouped data^{6,7}; the measurement-error lineage, which treats the rounding as error and deconvolves^{8,9}; and latent-variable models of the mechanism, led by the stochastic expectation-maximization of Groß and Rendtel¹⁰ and the mixture models of Wang and Heitjan⁵, which fit the reporting process and impute under the coarse-data framework¹¹. Each treats a different facet, and none appears to supply an exact account, in one domain, of what rounding does to the object density estimation manipulates, nor a statement of when the de-heaped density is recoverable.

This work supplies that account in the characteristic-function domain and draws estimators and an identifiability statement from it. The contributions are the following. First, an exact replica identity for the characteristic function of rounded data, of which Sheppard's correction is the leading moment term, with an explicit bound on the aliasing bias remaining after the box taper is divided out. Second, a grid-Nyquist identifiability limit, stated as a proposition and proved by a witnessing construction, saying which part of the density survives coarsening. Third, a de-heaping estimator with no user-set bandwidth, which deconvolves the known box kernel and sets its cutoff automatically. Fourth, a superposition variant carrying a smooth

base across the non-identifiable band, and a combined estimator selecting among the de-heaping, superposition and iterated forms from a band-capacity statistic. Fifth, direct readers of the grid width, the heaped fraction and the mixture of rounding grains, quantities the mechanism-fitting lineage obtains only through a full model fit, each with the regime in which it is valid fixed by the identifiability limit and mapped against known truth, together with the cost of estimating the grid rather than knowing it, measured end to end. Sixth, detection of the comb as a cyclic-group-matched atom by a fourth-order statistic in the spectral basis. Seventh, validation against three mainstream methods, which prior work on this problem did not benchmark against, on a selected synthetic set, on the full fifteen-density benchmark of Marron and Wand¹², and on two survey datasets, all computed in an accompanying open-source R library.

Relation to the companion spectral estimator

The estimator machinery builds on the spectral-decomposition density estimator of a companion preprint¹³ within the algebraic-diversity framework¹⁴. Three elements are inherited and are not claimed here: the spectral construction itself, the data-driven residue floor, and the largest-passing-group rule. Everything specific to heaping is new: the replica identity (2) and Proposition 1, the identifiability limit of Proposition 2, the box-deconvolution estimator and its band limit, the superposition and combined forms with the band-capacity statistic, the three comb readers, and the fourth-order detector. The companion preprint treats smoothing of a clean sample and contains no account of rounding, no identifiability statement and no comb.

The algebraic-diversity view of binned data

The estimators below derive from the algebraic-diversity framework^{13,14}, which exploits symmetry in data through group representations, and one fact ties it to this problem. Bin a sample into a probability vector p on M equal cells. The covariance of p averaged over the cyclic shift group, its Reynolds projection onto the commutant of the group action, is diagonalized by the discrete Fourier transform and has the squared empirical characteristic function $|\hat{\phi}(w_k)|^2$ as its eigenvalues. The algebraic spectrum of binned data is therefore the squared empirical characteristic function, and smoothing and de-heaping become operations on that spectrum. Supplementary Section S1 defines the framework in full, including the three derived notions used below: the *symmetry defect*, the *algebraic residue*, whose level read from the spectrum gives a data-driven noise floor, and the *largest-passing-group rule*.

Group discovery has also been studied in the supervised setting, where labels and the regular representation are available and the group is identified from many observations¹⁵. The setting here differs in all three respects: no labels, the covariance commutant of a binned sample rather than a regular representation, and a single observation. The two lines are complementary rather than competing.

Results

The characteristic function of heaped data

Let X have density f and characteristic function $\varphi(t) = \mathbb{E} e^{itX}$. The heaping grid width D is the spacing of the round values to which the data are snapped: $D = 5$ pounds when weights are recorded to the nearest five pounds, $D = 10$ when cigarette counts cluster on multiples of ten, $D = 1$ when an integer count rounds a continuous rate. Rounding to a known grid of width D gives $Y = D \text{round}(X/D)$, a lattice variable with masses

$$p_k = P(Y = kD) = \int_{(k-\frac{1}{2})D}^{(k+\frac{1}{2})D} f(x) dx = (f * b_D)(kD), \quad (1)$$

where b_D is the unit-height indicator of $[-D/2, D/2]$ (the width- D box) and $*$ denotes convolution, $(f * g)(x) = \int f(u)g(x-u)du$. The heaped masses are therefore samples, at spacing D , of f convolved with the box. By Poisson summation the characteristic function of the rounded variable is an aliased sum of box-tapered replicas,

$$\varphi_Y(w) = \sum_{m \in \mathbb{Z}} \varphi\left(w + \frac{2\pi m}{D}\right) \text{sinc}\left(\left(w + \frac{2\pi m}{D}\right) \frac{D}{2}\right), \quad \text{sinc}(u) = \frac{\sin u}{u}. \quad (2)$$

The $m = 0$ term $\varphi(w) \text{sinc}(wD/2)$ is the true characteristic function tapered by the box transfer function. The $m \neq 0$ terms are replicas at the reciprocal-grid frequency $2\pi/D$, the rounding comb, which lift the high-frequency noise floor and which a kernel estimate renders as spikes at the rounding marks. Expanding $1/\text{sinc}(wD/2) = 1 + w^2 D^2/24 + \dots$ recovers Sheppard's variance correction $-D^2/12$ as the leading moment, so (2) generalizes it from the second moment to the whole characteristic function. Numerically (2) reproduces the exact characteristic function of a rounded bimodal target to relative error 6.2×10^{-9} , the residual reflecting truncation of the replica sum rather than error in the identity, and 10^7 standard-normal draws rounded at $D = 0.4$ give an empirical variance of 1.0129 against the predicted 1.0133.

What the box division recovers, and what it leaves behind

Inside the grid-Nyquist band $|w| < \pi/D$ the $m = 0$ term dominates, and dividing the empirical characteristic function by $\text{sinc}(wD/2)$ removes the box taper and returns φ up to the contribution of the remaining replicas (Fig. 1a). That contribution is not zero, and (2) bounds it in one line.

Proposition 1. *For every w with $|w| < \pi/D$,*

$$\left| \frac{\varphi_Y(w)}{\text{sinc}(wD/2)} - \varphi(w) \right| \leq \sum_{m \neq 0} \left| \varphi\left(w + \frac{2\pi m}{D}\right) \right| \left| \frac{\text{sinc}(wD/2 + \pi m)}{\text{sinc}(wD/2)} \right|. \quad (3)$$

Well inside the band the replica terms are evaluated where φ has decayed, the right-hand side is negligible and the division is accurate; approaching the band edge the ratio of sinc factors grows and the bound degrades. At $w = \pi/D$ exactly the $m = 0$ and $m = -1$ tapers are equal, $\text{sinc}(\pm\pi/2) = 2/\pi$, so for a real symmetric target the division returns $2\text{Re } \varphi(\pi/D)$ rather than $\varphi(\pi/D)$ and the pointwise relative error is of order one. The useful statement is therefore (3) together with the observation that the bound is small over the interior of the band. Figure 1b reports that residual under both normalizations, since they differ by a factor of 19.9 at the edge and the choice governs how the panel reads. Divided pointwise by $|\varphi(w)|$, the quantity (3) bounds, the error is 1.000 there. Methods gives both normalizations across the band.

The identifiability limit

Beyond π/D the $m = \pm 1$ replicas overlap the signal and φ is not determined by φ_Y . The following makes that precise by exhibiting two densities that a rounding of width D cannot tell apart.

Proposition 2. *There exist densities $f_1 \neq f_2$ with identical lattice masses, $(f_1 * b_D)(kD) = (f_2 * b_D)(kD)$ for every $k \in \mathbb{Z}$, whose characteristic functions agree on $|w| < \pi/D$ and differ on $|w| > \pi/D$. The density is therefore not identifiable from the recorded values beyond the grid-Nyquist frequency.*

Both proofs are in Methods. Structure narrower than about $2D$ is aliased away and cannot be recovered from the recorded values alone, although recovery beyond the band under additional assumptions remains possible, through analyticity or the smoothness priors of the deconvolution literature¹⁶; the statement here is that the data alone do not determine it. The limit separates rounding from pure uniform measurement error, since convolution with a uniform density alone would leave φ recoverable to the first zero of the box factor at $2\pi/D$, while the lattice sampling aliases replicas into the band at π/D . It is the sampling and not the box that sets the limit, and it organizes everything that follows: it explains why the smooth base of the superposition carries the coarse regime, and why at the coarsest grids a method supplying a smooth prior retains an advantage no data-driven de-heaper can overcome.

A de-heaping estimator with no user-set bandwidth

The algebraic spectrum of the binned data is $|\hat{\varphi}(w_k)|^2$, and rounding is deconvolution against $\text{sinc}(wD/2)$ in that domain, tapered by the optimal linear gain against the noise floor. Within $|w| < \pi/D$ the estimator sets the deconvolved power $\hat{S}(w) = |\hat{\varphi}(w)|^2 / \text{sinc}^2(wD/2)$ and the amplified floor $\beta(w) = (1/n) / \text{sinc}^2(wD/2)$, applies the Wiener gain $g = (\hat{S} - \beta)_+ / ((\hat{S} - \beta)_+ + \beta)$, forms the net filter $g(w) / \text{sinc}(wD/2)$ on $\hat{\varphi}$, zeroes it outside the band, inverts, clips and renormalizes. The floor is amplified exactly where the box transfer function decays, which is the characteristic difficulty of deconvolution, and the residual-meets-floor rule sets the cutoff with no bandwidth chosen by hand. The band edge itself is not a tuned quantity; an ablation over cutoffs from half the band to the full band, in Methods and Supplementary Table S5, finds the error *monotone* and the full band best.

Superposition and the combined estimator

The band-limited estimator is empty beyond π/D , which is where Proposition 2 says a smooth base must carry the density, and the superposition variant supplies one. A Gaussian mixture is fit to jitter-dequantized data and its broad components form a smooth base that carries the non-identifiable band, the band-limited de-heaped residual computed on the leftover mass supplies the sharp in-band detail, and the two are added. Beyond π/D only the base contributes, so it fills the aliasing gap, and inside the band the residual sharpens it. An iterated form re-estimates the base from a resample of the current estimate for a few passes, the de-heaping analog of the imputation loop but using exact box deconvolution rather than cell imputation.

These forms win in different regimes and the identifiability limit says which. When the structure fits inside the band the de-heaping estimator is best and a base only adds bias; when it exceeds the band the superposition is needed, iterated at the coarsest grids. The combined estimator selects among them from a band-capacity statistic, the fraction of de-heaped in-band power in the outer quarter of the band, which is small when the spectral content has decayed inside the band and large when it is still strong at the grid-Nyquist edge. Low capacity selects the de-heaping estimator, moderate the superposition, high the iterated superposition. That single estimator is the one carried into the benchmark.

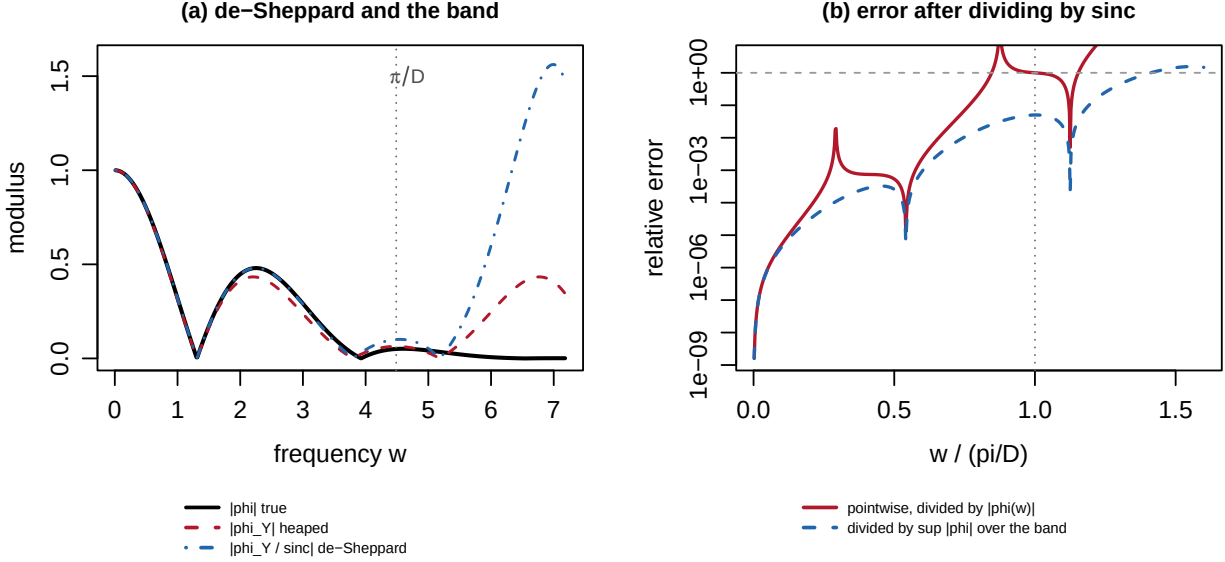


Figure 1. The identifiability band. (a) The heaped characteristic function $|\varphi_Y|$ departs from the truth $|\varphi|$; dividing by $\text{sinc}(wD/2)$ (de-Sheppard) removes the box taper and returns $|\varphi|$ over the interior of the grid-Nyquist band $|w| < \pi/D$, with the residual bounded by Proposition 1. (b) Relative error after de-Sheppard division against band fraction, under both normalizations: divided pointwise by $|\varphi(w)|$, which is what Proposition 1 bounds, and divided by the supremum of $|\varphi|$ over the band. The pointwise error is small well inside the band and reaches 1.000 at the edge; the sup-norm curve reaches 0.050 there. The vertical axis is capped because the pointwise ratio diverges at the zeros of $|\varphi|$. Bimodal target, $D = 0.70$, replica sum truncated at $|m| = 8$.

What tuning-free means here

The estimator requires no bandwidth from the user, and that is the property the title claims. It claims no more than that. The construction contains several constants, listed in Methods, that are fixed once and held fixed across every experiment rather than eliminated. None is set per dataset and none is tuned at run time, so the estimator is free of a user-set bandwidth in the sense that matters for practice, while the tuning is frozen rather than removed. Whether such constants carry a hidden dependence on the targets they are reported against is a measurable question. A joint sweep over the two band-capacity thresholds, 0.06 and 0.14, in Methods and Supplementary Table S4, places the low threshold in the middle of an exactly flat plateau and moves the mean error by about 1.9 percent across the high threshold, so the results of Table 1 do not turn on either value. A slightly better pair exists on the sweep and is not adopted, for the reason given there.

Benchmark against the dedicated literature

The method evaluated is a single estimator, the combined estimator, built from the de-heaping estimator and the superposition. Table 1 separates those two components from the combined estimator that deploys them and compares it against three external methods and the naive kernel, all computed in the accompanying R library: fixed-bandwidth box deconvolution (deconv)⁸, the `Kernelheaping` stochastic expectation-maximization of Groß and Rendtel¹⁰ (SEM), and a base-R replica of Heitjan-Rubin multiple imputation¹¹, labeled `imput`. Integrated squared error against the true density is reported as the grid coarsens, for $n = 4000$ over 50 seeds, on four Gaussian-mixture targets spanning smooth unimodal, bimodal, spike on a broad base, and strongly skewed shapes. Two are members 1 and 4 of the normal-mixture benchmark of Marron and Wand¹² and two are custom, and all fifteen members are evaluated on the same grid, seeds and estimators in Supplementary Table S2. Ties are decided by a per-seed paired difference within two paired standard errors, and Supplementary Table S3 reports those differences so every bolding decision can be checked.

The combined estimator is best of the corrected methods, or tied with the best, in 12 of the 16 cells. It inherits the de-heaping estimator's advantage on smooth densities at fine and moderate grids, where its automatic cutoff outperforms a hand-set bandwidth, and the superposition's advantage on multi-scale densities at moderate grids. At bimodal $D = 1.00$ the SEM and combined errors, 7.099 against 7.596, are statistically indistinguishable, as are the strongly skewed values at

Table 1. Integrated squared error ($\times 10^3$) against the true density as the heaping grid D coarsens. Mean over 50 seeds with the seed-to-seed standard deviation in parentheses, $n = 4000$, computed in the accompanying R library on a grid of $M = 2048$ cells of width $\Delta x = 0.009766$. The four targets are a selected set. Two are members 1 and 4 of the normal-mixture benchmark of Marron and Wand; the bimodal and strongly skewed targets are custom and are defined in Supplementary Table S1. All fifteen members of that benchmark are evaluated in Supplementary Table S2. The three external methods are deconv, SEM and imput. The proposed method is the combined estimator, whose two constituent components are set off by vertical rules. In each row the best of the four heaping-corrected values is bold, together with any corrected value whose per-seed paired difference from the best is within two paired standard errors; those paired differences and their standard errors are reported in Supplementary Table S3. The naive kernel is the uncorrected reference. The combined estimator is the best or tied with the best in 12 of the 16 cells.

density	D	naive	deconv	SEM	imput	de-heap	super	combined
Gaussian	0.25	0.384 (0.19)	0.391 (0.19)	0.486 (0.22)	0.358 (0.19)	0.623 (0.46)	0.541 (0.46)	0.623 (0.46)
	0.50	1.447 (0.15)	0.401 (0.18)	0.444 (0.20)	0.358 (0.19)	0.290 (0.18)	0.250 (0.21)	0.290 (0.18)
	1.00	101.3 (4.70)	0.266 (0.17)	0.427 (0.18)	1.416 (0.34)	0.234 (0.16)	0.646 (0.30)	0.234 (0.16)
	1.50	243.2 (7.86)	1.545 (0.16)	1.595 (0.31)	5.378 (0.63)	1.408 (0.12)	7.438 (2.69)	1.408 (0.12)
Bimodal	0.25	4.015 (0.59)	3.590 (0.57)	3.470 (1.09)	3.594 (0.55)	0.835 (0.40)	0.736 (0.40)	0.835 (0.40)
	0.50	5.912 (0.77)	4.007 (0.72)	3.544 (1.35)	5.040 (0.72)	0.801 (0.34)	1.379 (0.53)	0.801 (0.34)
	1.00	48.746 (1.13)	12.535 (0.43)	7.099 (0.69)	14.014 (1.03)	9.904 (0.24)	9.862 (1.81)	7.596 (2.53)
	1.50	124.9 (1.42)	72.136 (0.71)	19.481 (1.48)	29.604 (1.57)	74.415 (0.88)	35.783 (12.67)	68.522 (13.75)
Kurtotic	0.25	71.005 (3.75)	65.423 (3.55)	21.149 (4.50)	76.356 (3.35)	32.038 (1.52)	32.123 (1.53)	32.123 (1.53)
	0.50	36.314 (3.29)	123.5 (0.67)	69.581 (2.44)	93.375 (2.13)	128.7 (1.12)	46.744 (2.62)	41.999 (5.23)
	1.00	68.891 (1.15)	213.5 (0.46)	167.1 (5.76)	173.2 (2.01)	213.7 (0.49)	145.5 (2.88)	145.5 (2.88)
	1.50	126.3 (2.49)	251.6 (0.24)	205.5 (5.64)	229.2 (2.11)	251.5 (0.22)	192.2 (2.66)	192.2 (2.66)
Strongly skewed	0.25	4.678 (0.94)	3.759 (0.89)	3.284 (0.80)	5.002 (0.90)	1.217 (0.53)	1.451 (0.53)	1.217 (0.53)
	0.50	19.955 (1.38)	7.901 (0.95)	7.782 (1.13)	9.652 (1.31)	6.112 (0.53)	6.210 (1.01)	6.112 (0.53)
	1.00	301.6 (7.55)	47.980 (1.27)	45.515 (2.79)	65.211 (2.84)	45.921 (1.25)	52.266 (3.28)	45.921 (1.25)
	1.50	281.4 (12.11)	71.566 (0.63)	21.940 (2.35)	38.929 (1.04)	71.681 (0.67)	50.702 (2.74)	50.702 (2.74)

$D = 1.00$.

The four cells where the combined estimator is neither best nor tied are of two kinds. At the finest Gaussian grid the heaping is negligible and the correction adds variance rather than removing bias: the external corrected methods sit near the no-heaping error and the proposed method does not, giving 0.623 against a naive 0.384, a factor of 1.62, with a paired difference of +0.239 and paired standard error 0.070. The same pattern appears on the NHANES weights at a five-pound grid in Table 2. The other three are the coarsest multimodal grids and the narrowly aliased kurtotic spike at the finest grid, where the sharp structure lies beyond π/D , Proposition 2 says no data-driven de-heaper recovers it, and only the SEM imputation's smoothness prior helps, by assuming rather than recovering.

On the full fifteen-density benchmark of Supplementary Table S2 the combined estimator is best or tied in 37 of 60 cells, and the pattern across grids is the one Proposition 2 predicts: 12 of 15 at $D = 0.25$, 12 of 15 at $D = 0.5$, 8 of 15 at $D = 1.0$ and 5 of 15 at $D = 1.5$. The decline is monotone in the grid width. The least favorable case anywhere is the outlier density, member 5, where at $D \geq 0.5$ every corrected method is worse than doing nothing, for the same reason as the kurtotic rows.

Sweeping one target across 7 grids shows the same boundary in the estimator's own error curve (Fig. 2). The bimodal target has $\sigma_{\min} = 0.50$, so those grids span $D/\sigma_{\min}$ from 0.2 to 3.0. The combined estimator is best of the corrected methods or tied with the best at 5 of them, every one with the ratio at or below 2.0, and at the remaining 2, where it is 2.5 and above, costs a factor of 3.6 to 6.3 against the SEM imputation. It still improves on the naive estimate there, so what the coarse grids remove is its standing against the alternatives rather than its usefulness. That threshold is the 2.5 the heaped-fraction reader gives, reached by a different route; Methods reports which part of the combination fails, and a cross-check of this sweep against Table 1.

Reading the grid, the heaped fraction, and mixed grains from the comb

When D is unknown it is read from the first replica: the leading tooth of $|\hat{\phi}(w)|^2$ above the noise band sits at $w = 2\pi/D$, so $\hat{D} = 2\pi/w_{\text{peak}}$. The reader admits two modes, a verifying mode in which the tooth is sought within a window around an expected grid and a blind mode in which the first strong tooth beyond the signal lobe is taken. The two are not equivalent. Over the targets of Table 1 at three grids the verifying mode returns the imposed grid in 12 of 12 cases and the blind mode in 0, so a blind reading is a candidate rather than a recovered grid, and a recovery obtained in the verifying mode is a confirmation of an expected grid rather than a detection (Supplementary Table S9).

Estimating the grid and then using it is a further question, and running the reconstruction from each estimate separates

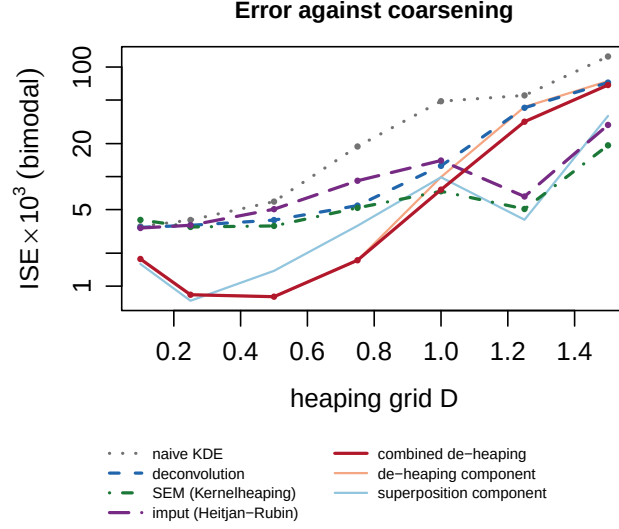


Figure 2. Integrated squared error against the heaping grid D on the bimodal target, $n = 4000$, 50 seeds, over 7 grids from $D = 0.10$ to $D = 1.50$, carrying the comparison set of Table 1 together with the two components of the combined estimator. The naive estimate grows by a factor of 35.6 across the sweep. The combined estimator is best of the corrected methods or tied with the best while $D/\sigma_{\min}$ stays at or below 2.0, and loses that standing at 2.5, which for this target is $D = 1.25$.

the reader’s error from the estimator’s (Supplementary Table S10, with the construction in Methods). The verifying estimate reproduces the oracle result, which is the Table 1 column, in 12 of the sixteen cells. The blind estimate returns nothing usable in 7 cells, and where it does return something the end-to-end error tracks how far it falls from the imposed grid: within about 25 percent the reconstruction costs a factor of 0.97 to 2.00 against the oracle, and beyond that a factor of 12.3 to 139.7. The de-heaping therefore tolerates a moderate error in D , and the failure of an automatic pipeline is in the reader rather than in the estimator.

Under partial heaping, where the law is $(1 - p)f + pf_{\text{heaped}}$, only the heaped fraction feeds the comb and the leading tooth has height approximately p^2 over the floor, so $\hat{p} = \sqrt{S_{\text{peak}} - \hat{b}}$. What that recovers is again set by Proposition 2, through the ratio $D/\sigma_{\min}$ of the grid width to the smallest scale in the target. At or below about 2.5 the recovery is monotone in *every* one of the 6 cells tested and linear to a coefficient of determination of at least 0.999; above it monotonicity fails in 4 of 6 cells and the reader returns as much as 0.358 on unheaped data (Supplementary Table S11). The attenuation is not one constant across targets and grids, so the reader is a monotone indicator of p within the fine-grid regime rather than a calibrated estimate, and a recovery measured in one cell does not generalize to another.

Naturally heaped counts often mix grains, and integer counts anchor the unit grain. A class of reports rounded to grain g is a lattice variable with spacing g , so its characteristic function is exactly one at every multiple of $2\pi/g$. At a center of a coarser grain that is not a center of any finer one, the coarser class contributes its full weight while each finer class contributes the Fourier transform of its residue-class distribution modulo the coarser grain, which vanishes when the base is smooth at that scale; the Möbius differences of the amplitudes at those exclusive centers are then the grain weights. When the base is sharp at that scale the finer classes alias into the coarser centers and the reading is biased, most on the coarsest grain, which is Proposition 2 applied to the mixture. The regime is set by κ , the base standard deviation over the coarsest grain: on a known-truth mixture the reader recovers every weight to within 4.1 percent once κ exceeds about three, and recovers the unit share, hence the heaped fraction, at every κ tested (Supplementary Table S8).

The comb as a group-matched atom in the spectral basis

The comb is a cyclic-lattice group-structured component, so it should be detectable by the largest-passing-group rule. Applying that rule to the covariance of the binned histogram fails, because the finite-support histogram is not shift-invariant under the grid period. The correct basis is the spectral one. The characteristic function of a lattice variable is periodic with period

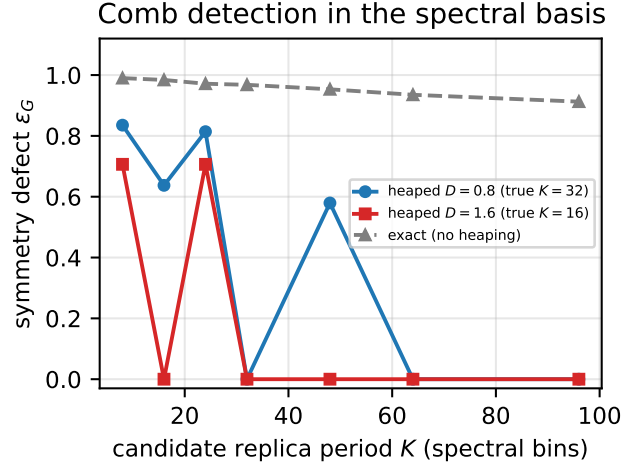


Figure 3. Higher-order comb detection in the spectral basis. The symmetry defect of the shift-by- K quotient group, evaluated on the covariance of the power spectrum (a fourth-order statistic of the data), drops to zero at the true replica period K and its multiples for heaped data, and stays high for all periods on exact data. The smallest period with low defect, the largest passing cyclic subgroup, is the detected grid.

$2\pi/D$, so the power spectrum is a full-support periodic sequence invariant under a shift by the replica period $K = M/P$ in the frequency index, where $P = D/\Delta x$. The covariance of that spectrum is a fourth-order statistic, and applying the group selector to it recovers the grid: the symmetry defect of the shift-by- K quotient group drops toward zero at the true replica period and its multiples, and the smallest period with low defect is the detected grid (Fig. 3). The detector abstains on exact unheaped data, where no non-trivial period passes.

With the default settings of the accompanying library, stated in Methods, the detector is conservative rather than accurate on demand: over the sixteen cells of Table 1 it fires in a majority of replicates in 0 of them. Of the two automatic routes to a grid it is nonetheless the more promising, because it abstains rather than guessing. Where it does fire and the reconstruction is run from its estimate, the result is within about a factor of two of the oracle in 7 of the 8 cells in which it ever fires and better than the oracle in 3, against a leading-tooth reader that rarely abstains and is often far wrong (Supplementary Tables S9 and S10, with the counts in Methods).

Validation on NHANES survey microdata

Table 2 reports a controlled study on the measured adult weights of the NHANES 2017–2018 body-measures file¹⁷, $n = 5185$, computed in the R library against all three external methods. A known grid of width D is imposed by rounding and the density is estimated from the rounded values, against a Gaussian kernel estimate of the same unrounded measurements as the no-heaping reference, so the comparison isolates the response to coarsening; Methods gives the construction in full. Because that reference is itself a kernel estimate at one bandwidth, error against it partly rewards resemblance to that smoothing rather than recovery, and the ranking of the methods depends on that bandwidth. Sweeping the reference bandwidth reverses it: the combined estimator is first at 4 of the five grids at half the Silverman bandwidth and third or fourth at most grids at the Silverman bandwidth and above (Supplementary Table S7, with the sweep in Methods). The Table 2 ranking is therefore specific to its reference bandwidth.

Below about half the kernel bandwidth the heaping is invisible and the estimators agree. As the grid coarsens the naive estimate develops spurious modes and degrades by several orders of magnitude (Fig. 4), while the combined estimator and the three external methods stay within a small multiple of the no-heaping error. At the finest grid the negligible-heaping effect described for Table 1 reappears: the external corrected methods track the reference closely and the combined estimator does not. Every cell of Table 2 is emitted at four decimals from the stored study, so the table and the panel (A) axis now carry the same quantity.

A naturally heaped variable with no continuous ground truth is the self-reported number of cigarettes smoked per day, which mixes several grains. Among the 1019 adult smokers reporting a daily count, 57 percent gave a multiple of five, 42 percent a multiple of ten and 22 percent a multiple of twenty, so terminal digit zero accounts for 42 percent of reports against the 10 percent expected under no digit preference⁵. Those shares are direct counts of the data. The lattice reader, anchored on the unit grid, fires with a statistic of $\lambda = 3.80$, so the mixture is detected; λ is a log ratio of tooth power to a robust spectral

Table 2. NHANES measured adult weights rounded to grid D (lb): integrated squared error ($\times 10^3$) against the no-heaping kernel reference, $n = 5185$, computed in the R library on a grid of $M = 4096$ cells at a reference bandwidth of 9.681 lb. $\hat{D}$ is the grid located from the comb in the verifying mode, with the imposed grid supplied as the expected one, and is a confirmation rather than a blind detection. Every value is emitted from `nhanes_v10_7.rds`.

D (lb)	naive	deconv	SEM	imput	combined	$\hat{D}$
5	0.0001	0.0001	0.0014	0.0001	0.0536	4.98
10	0.0003	0.0003	0.0010	0.0004	0.0033	10.06
20	0.0035	0.0024	0.0044	0.0015	0.0070	20.11
30	0.2160	0.0036	0.0018	0.0028	0.0053	30.17
40	1.1635	0.0042	0.0034	0.0160	0.0062	38.78

Table 3. Berlin resident-register age distribution regrouped to a uniform grain of D years (from a native mixed grain of one to fifteen years): integrated squared error ($\times 10^3$) against the native-resolution reference density, pooled over 447 areas and 3.61 million residents, computed in the R library on a grid of $M = 2048$ cells. $\hat{D}$ is the grain located from the comb in the verifying mode, with the imposed grain supplied as the expected one.

D (yr)	naive	deconv	SEM	imput	combined	$\hat{D}$
5	0.006	0.003	0.152	0.007	0.015	5.00
10	0.984	0.024	0.088	0.404	0.040	10.00
15	5.331	0.273	0.160	1.299	0.304	14.55
20	11.189	0.257	0.245	2.013	0.254	20.00

floor, defined in Methods, and is descriptive rather than calibrated.

The dequantized counts have a standard deviation of about nine against a coarsest grain of twenty, so $\kappa = 0.44$, well inside the regime in which the reader is biased on the rounding grains. The comb therefore supports two quantities on these data and not a third. The heaped fraction is 0.467 with bootstrap standard error 0.019 over 1000 resamples, which the sweep supports as a measurement at this κ . The split of that fraction among the grains, 0.533, 0.176, 0.167, 0.124 for the unit share and the five, ten and twenty grains with standard errors 0.019, 0.024, 0.024, 0.008, is reported for completeness and not as a measurement, and Supplementary Fig. S1 illustrates such a decomposition on these counts in the same spirit. Recovering grain-by-grain weights on data of this shape would need either a base smooth at the twenty grain or a model of the base, and the second is the mechanism-fitting route this work does not take.

Validation on the Berlin resident register

A second survey dataset exercises a different heaping mechanism, interval grouping rather than digit-preference rounding, on the dataset used to develop the dedicated `Kernelheaping` method¹⁰. The Berlin resident-registration matrix of December 2015¹⁸ records each of 3,610,156 residents across 447 planning areas with age given only within a known interval, of widths one to fifteen years, and grouping is the limiting case of heaping in which every value in a cell is collapsed to the cell. Age-band counts are pooled over the areas and dequantized within their intervals to give a native-resolution sample whose kernel estimate is the reference, and coarser uniform grids of D years are imposed by regrouping; Methods gives the construction. Placing values uniformly within a cell is the operation the imputation baselines themselves perform, so the construction favors those baselines at fine grains rather than the proposed method.

Table 3 reports the integrated squared error against that reference. As the grouping coarsens from five to twenty years the naive estimate degrades by more than three orders of magnitude, developing a staircase at the interval edges (Fig. 5), while the combined estimator and the external methods stay bounded and the imposed grain is located to within the grid resolution. This is the interval-grouping counterpart of the NHANES study, on real administrative data.

Two statements need care. At the finest grains the combined estimator is far more accurate than the `Kernelheaping` SEM on its own development dataset, 0.015 against 0.152 at five years and 0.040 against 0.088 at ten, because the stochastic-imputation smoothing oversmooths fine-grain structure that band-limited de-heaping preserves. It is not, however, the most accurate method there, since deconvolution is lower at both grains and the naive estimate is lower at five years, and it does not stay within a small multiple of its finest-grain error, rising from 0.015 at five years to 0.304 at fifteen. At the coarsest grains SEM regains a small edge, the same identifiability boundary as before.

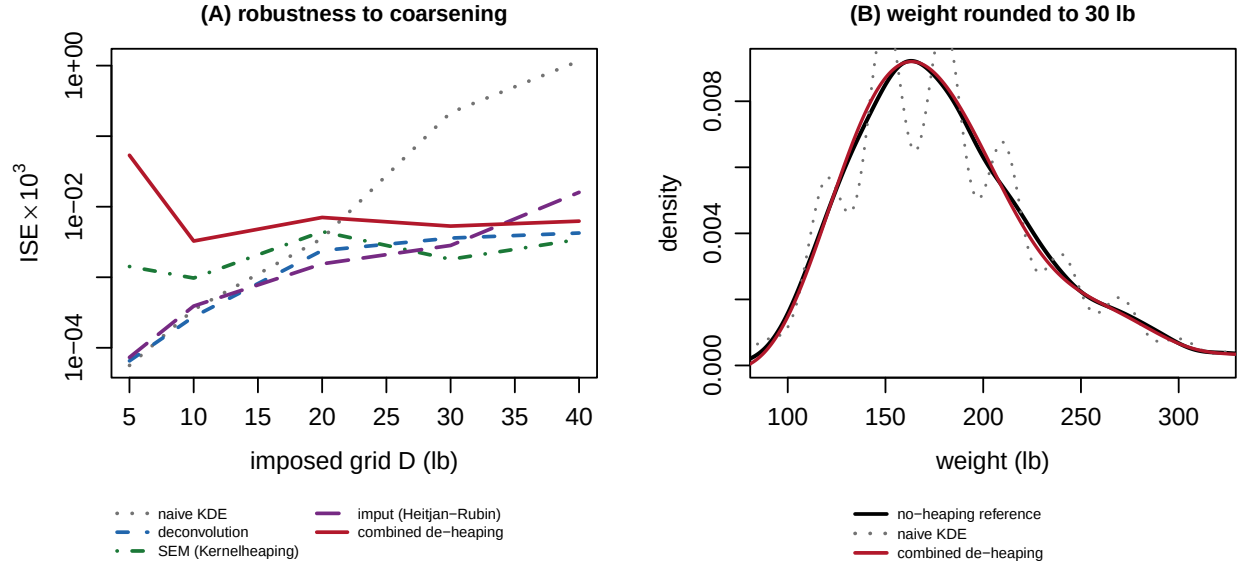


Figure 4. Controlled heaping on NHANES measured weights, computed in the R library. (A) Integrated squared error against the no-heaping density as the imposed grid coarsens. The vertical axis is logarithmic in the same $ISE \times 10^3$ that Table 2 prints, so a tick on the panel is a column entry in the table. (B) At a 30 lb grid the naive estimate oscillates at the rounding marks while the combined de-heaping estimator tracks the no-heaping density.

Discussion

Heaping has an exact characteristic-function description as box convolution followed by lattice sampling, and the identifiability limit that follows from it is the organizing principle of everything above. It draws the line between what a de-heaper can recover and what only a smooth prior can supply, it explains why the superposition is needed in the coarse regime, and it bounds every data-driven method together at the coarsest grids, where the stochastic-imputation estimator's residual advantage is an advantage of assumption rather than of recovery. The fifteen-density benchmark in the supplement shows that boundary as a monotone decline in relative performance as the grid coarsens.

Several limitations bound the result. When the heaping grid coincides with the scale of genuine structure, no blind procedure separates the comb from the density, and the same ratio bounds every reader the comb supports: the grid, the heaped fraction and the grain mixture are each recoverable on one side of it and not the other, which is Proposition 2 appearing three times. The estimator is subject to the same boundary and not merely the readers that feed it, since its advantage over the alternatives ends at the same ratio (Fig. 2). Estimating the grid and then using it is measured here rather than assumed, and the leading-tooth reader is not accurate enough to carry an automatic pipeline, although the reconstruction tolerates a grid error of about 25 percent. The de-heaping is exact only for mass-preserving, locally symmetric rounding; systematic rounding in one direction leaves a bias the recorded values cannot reveal. Past the grid-Nyquist frequency any recovery reflects the estimator's smoothness prior rather than the data. When the heaping is negligible relative to the smoothing scale the correction adds variance without removing bias, and on the evidence of Tables 1 and 2 an uncorrected kernel estimate is preferable there. An automatic rule for when to abstain would remove that regime. Three were tried in all 76 cells at 50 seeds and none pays for itself, as Methods reports, so none is adopted and the regime is reported as one in which the method should not be used rather than one it detects. The detector treats the comb at fourth order, and a fuller treatment across higher orders, together with a blind estimate of the box kernel when the grid must be learned jointly with the density, are the natural next steps. Because the estimators are differentiable almost everywhere, generative constructions can optimize against the recovered density directly.

When de-heaping is possible had been addressed in the deconvolution literature through ill-posedness and identifiability results, and the periodicity of the characteristic function of a lattice variable is classical. What appears to be new is that these turn the boundary into an operational criterion, computable from the recorded values, for when de-heaping should be attempted and for what the comb can be asked to report.

Against the dedicated measurement-error and latent-variable methods on a selected synthetic set, on the full Marron-Wand

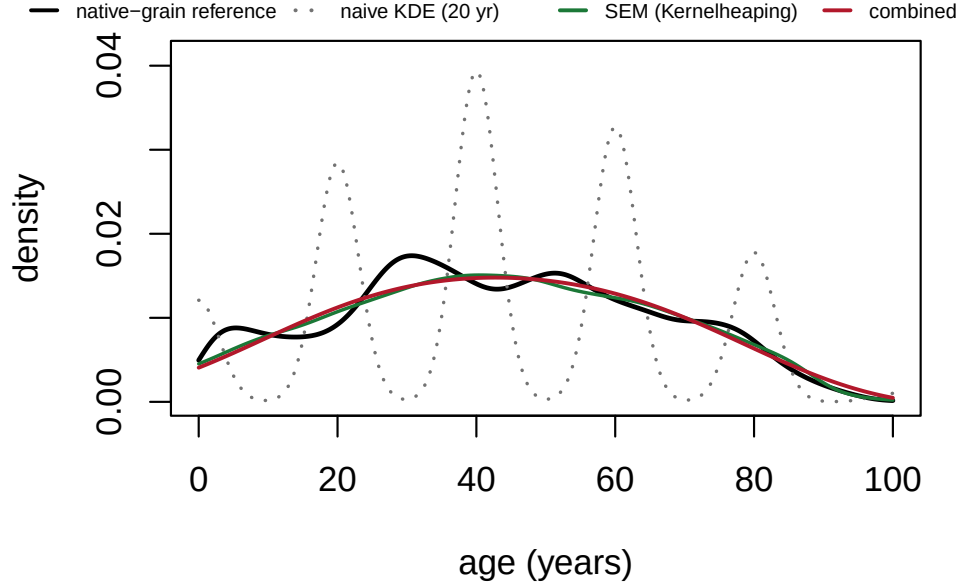


Figure 5. Berlin resident-register age density regrouped to a twenty-year grain. The naive estimate carries a staircase from the coarse grouping; the SEM and combined de-heaping estimators recover the smooth native-resolution age density.

benchmark and on two survey datasets, the single combined estimator is competitive or better in the majority of regimes. Where it is beaten at the coarsest multimodal grids the identifiability limit shows recovery is not available from the data alone, so the advantage a mechanism model holds there is one of assumption rather than information, and where it is beaten at the finest grids the heaping is negligible and the correction is not warranted. The result is a heaped-data density estimator free of a user-set bandwidth, free of a rounding-mechanism model, and explicit about the regimes in which it should not be used.

Methods

Proofs

Proposition 1. Divide (2) by $\text{sinc}(wD/2)$, which is nonzero for $|w| < \pi/D$, subtract $\varphi(w)$ to cancel the $m = 0$ term, and apply the triangle inequality to what remains. Writing $\text{sinc}((w + 2\pi m/D)D/2) = \text{sinc}(wD/2 + \pi m)$ gives (3).

Proposition 2. Let ψ be real, smooth, supported in $(\pi/D, 3\pi/D)$ and symmetric about $2\pi/D$, and set $\hat{g}(w) = |w| \psi(|w|)$, so g is real and even, integrates to zero, and $\hat{g}$ vanishes on $|w| < \pi/D$. By (2) the lattice masses of g have discrete-time Fourier transform $\sum_m \hat{g}(w + 2\pi m/D) \text{sinc}(wD/2 + \pi m)$. For $|w| < \pi/D$ only $m = \pm 1$ contribute, giving $\hat{g}(a) \text{sinc}(aD/2) + \hat{g}(-b) \text{sinc}(bD/2)$ with $a = w + 2\pi/D$, $b = 2\pi/D - w$, $a + b = 4\pi/D$. With $\theta = aD/2 \in (\pi/2, 3\pi/2)$ one has $\text{sinc}(aD/2) = \sin \theta / \theta$ and $\text{sinc}(bD/2) = -\sin \theta / (2\pi - \theta)$, and $\hat{g}$ real and even gives $\hat{g}(-b) = \hat{g}(b)$. The sum is $\sin \theta [a\psi(a)/\theta - b\psi(b)/(2\pi - \theta)]$, and since $a/\theta = b/(2\pi - \theta) = 2/D$ it vanishes identically when $\psi(a) = \psi(b)$, which is the assumed symmetry. Hence every lattice mass of g is zero. Take f_1 any density bounded below by $\varepsilon > 0$ on the support of g and $f_2 = f_1 + \delta g$ with δ small enough that $f_2 \geq 0$. Both integrate to one and have the same lattice masses, they agree on $|w| < \pi/D$ because $\hat{g}$ vanishes there, and they differ where $\hat{g}$ does not. $\square$

Spectral construction and estimators

The sample is binned to a probability vector p on M cells. As summarized above, the cyclic group-averaged covariance of p has the squared empirical characteristic function $|\hat{\varphi}(w_k)|^2$ as its eigenvalue spectrum, computed by the fast Fourier transform. All spectral filters are real, even functions of the angular frequency $w_k = 2\pi k/(M\Delta x)$. The naive estimator applies a Gaussian taper at the Silverman bandwidth. Two noise floors appear in this work and are not interchangeable. The residue floor is the motivating construction and is what the companion estimator uses on unrounded data. On coarsely rounded data it is not usable, for the reason recorded in the Results, so the de-heaping estimator uses the parametric floor $(1/n)/\text{sinc}^2(wD/2)$, the sampling floor amplified by the same box transfer function the estimator divides out. The residue floor appears only in the heaped-fraction reader, where the spectrum is not divided by the box factor and the floor search remains well posed.

The reason the residue floor fails on coarsely rounded data is that such a sample is effectively discrete, so its empirical characteristic function is quasi-periodic and never descends to a floor, the median floor is inflated by the comb, and the estimate becomes degenerate; the band limit and the parametric floor replace the floor search in that regime. The de-heaping estimator applies the box-deconvolution Wiener filter of the Results within the band $|w| < \pi/D$. The superposition estimator fits a Bayesian-information-criterion Gaussian mixture of up to six components to jitter-dequantized data $y + U(-D/2, D/2)$, forms a smooth base from its broad components, those whose standard deviation is at least 1.5 times the Silverman bandwidth of the jittered sample, computes the de-heaped residual on the leftover binned mass, and adds the two. The iterated form re-estimates the base from a resample of the current estimate for three passes. The combined estimator computes the band-capacity statistic, the fraction of de-heaped in-band power in the outer quarter of $|w| < \pi/D$, and selects the de-heaping estimator below a low threshold of 0.06, the superposition between the thresholds, and the iterated superposition above a high threshold of 0.14. The two thresholds are fixed across all experiments. The constants referred to in the Results as frozen rather than eliminated are these two band-capacity thresholds, the broad-component rule at 1.5 times the Silverman bandwidth of the jittered sample, the mixture cap of six components, the three passes of the iterated form, the detector's defect band of 0.35, its spread requirement of fifteen percent, and the lattice reader's firing rule at four times the floor amplitude. None is set per dataset and none is tuned at run time.

Threshold sweep, cutoff ablation, and abstention rules

The two band-capacity thresholds were swept jointly over the sixteen cells of Table 1 at 20 replicates (Supplementary Table S4). The mean error at the adopted pair (0.06, 0.14) is $37.22 (\times 10^3)$. Setting the low threshold to 0.04, 0.06, 0.08 gives results identical to every printed digit, because no cell's band-capacity statistic falls in that interval, and across high thresholds from 0.10 to 0.20 the mean error moves between 36.93 and 37.65, a spread of 1.9 percent of that value. A threshold sitting in a gap of the empirical distribution of its statistic cannot have been tuned to these data in any way that matters. The best cell of the sweep, 35.50 at (0.04, 0.25), is 4.6 percent better than the adopted pair and is not used, since choosing it would tune a constant on the cells the table reports.

The band cutoff of the de-heaping estimator was placed at fractions from 0.5 to 1.0 of π/D on the same cells (Supplementary Table S5). The mean error is *monotone* in the fraction, falling from 94.7 at half the band to 52.4 at the full band, so the full band is retained.

Three abstention rules, each returning the naive estimate when a gate fires, were run in every cell of Table 1 and Supplementary Table S2, 76 cells at 50 seeds: a gate on the fourth-order comb detector, a gate on the ratio of grid width to Silverman bandwidth at a multiplier of 0.5, and a gate on the box taper at the frequency where the spectrum meets the floor. Measured by the number of Table 1 cells in which the combined estimator is best or tied, the unmodified estimator scores 12 of 16, the detector gate 3, the scale gate 12 because it never fires at that multiplier, and the taper gate 6. Every gate catches the cell it was aimed at, the taper gate returning exactly the naive 0.384 at the Gaussian $D = 0.25$ cell, and every gate costs more elsewhere than it saves there, the detector gate adding $1328.5 (\times 10^3)$ to the total error over the 76 cells. Each also introduces a constant of the kind the tuning-free claim is meant to avoid, so none is adopted.

Evaluation grids and what each estimator is given

Every experiment is stated here with its binning, because the spectral resolution, the noise floor, and the grid readout all depend on it. The synthetic benchmark of Table 1 and Supplementary Table S2 uses $M = 2048$ cells spanning $[-10, 10)$, so $\Delta x = 0.009766$. The NHANES study of Table 2 uses $M = 4096$ cells spanning the measured weight range extended by 40 pounds at each end. Measured weights in kilograms are converted to pounds and retained for respondents aged twenty and over with a non-missing measurement; the no-heaping reference is a Gaussian kernel estimate of those unrounded values at the Silverman bandwidth on the same evaluation grid. The Berlin study of Table 3 uses $M = 2048$ cells spanning $[-25, 135)$ years. Age-band counts are pooled over all 447 areas, a sample of 40,000 values is drawn from the pooled counts in proportion to their totals under the seed of record, with each drawn value placed uniformly at random within its native interval, and the reference density is a Gaussian kernel estimate of that dequantized sample at the Silverman bandwidth. The register covers 3.61 million residents and the reference is built from that draw rather than from every record, so the comparison is limited by the draw rather than by the register. Coarser uniform grids are imposed by regrouping the same pooled counts. The grid readout of the comb is $\hat{D} = M\Delta x/K$ for a detected replica period K in frequency index.

In the synthetic benchmark and in both controlled real-data studies the grid width D is imposed by the experiment and the true value of D is supplied to every estimator that takes it, the proposed one and all three external baselines alike. No estimator in those tables uses $\hat{D}$ from the comb, and the $\hat{D}$ columns of Table 2 and Table 3 report the reader's output alongside the imposed value rather than feeding it back into the estimate. The comparison is therefore like-for-like with respect to the information supplied. How the method performs when D must be estimated first is measured separately, below.

External baselines

Three external methods are compared. The measurement-error deconvolution baseline (deconv) divides the empirical characteristic function by $\text{sinc}(wD/2)$ within the band with a fixed Silverman Gaussian cutoff and a standard ridge cap, the rounding-as-uniform-error lineage⁸. Because a data-driven bandwidth selector is available for this setting¹⁶, a fixed bandwidth could understate the baseline, and the question is settled by rerunning it at two further bandwidths on the sixteen cells of Table 1 at 20 replicates (Supplementary Table S6): least-squares cross-validation, which a practitioner can compute, and an oracle bandwidth minimizing the true error, which none can. Cross-validation is worse than the fixed Silverman bandwidth in 9 of the 16 cells, so the fixed bandwidth is not a handicap imposed on the baseline, although where cross-validation helps it helps substantially, taking the bimodal $D = 0.25$ cell from 3.418 to 0.891. Against the combined estimator's Table 1 values, the combined estimator is better than the cross-validated baseline in 13 of 16 cells and better than the oracle in 8; in the kurtotic cells at $D = 0.5$ and $D = 1.5$ the oracle reaches 123.4 and 251.4 against 41.999 and 192.2, so no bandwidth rescues the baseline there. In 2 cells the oracle search stopped at the lower end of its bandwidth grid. The SEM baseline is the real Kernelheaping package of Groß and Rendtel^{10,19}, called with its default burn-in and sample counts and interpolated to the common grid. The input baseline is a base-R replica of the multiple-imputation approach of Heitjan and Rubin for coarse data¹¹: a pilot density is fit to jitter-dequantized values, then in each of eight independent imputations every heaped value is imputed within its rounding cell in proportion to the pilot and a kernel estimate is formed, and the imputations are averaged. No standalone kernel-density package for that approach exists, so this replica stands in, and it has not been validated against a published implementation. It should be read as a reasonable rendering of the method rather than as a certified one.

All estimators, the proposed methods and the three baselines alike, are implemented in an accompanying base-R library, so every table is computed in one language. The R and Python de-heaping estimators agree to 3×10^{-16} on a shared sample. That figure establishes that two implementations by one author transcribe the same specification, and it does not independently certify that specification. Integrated squared error is computed exactly against the closed-form Gaussian-mixture targets on a common grid, against a Gaussian kernel estimate of the unrounded values for the NHANES study, and against a Gaussian kernel estimate of the native-resolution dequantized sample for the Berlin study. The kurtotic rows need care. At $D \geq 0.5$ the naive kernel attains the lowest raw error of any method, because the spike concentrates mass at the rounding scale and the comb the naive estimate renders happens to sit on top of genuine structure, so the error metric rewards a coincidence of grid and structure rather than a recovery. The corrected methods, which cannot distinguish that structure from the comb they remove, are the like-for-like comparison set. Since 76 cells are reported in total, the tie decisions should be read with the number of comparisons in mind, and a few borderline calls would be expected to fall either way by chance. The four cells of Table 1 where the combined estimator is neither best nor tied are not borderline, the smallest paired gap being 3.8 paired standard errors and the largest 60.2.

Comb readers and the spectral detector

The grid is read as $\hat{D} = 2\pi/w_{\text{peak}}$ from the leading replica tooth of $|\hat{\phi}|^2$ above the noise band, and the heaped fraction as $\hat{p} = (S_{\text{peak}} - \hat{b})^{1/2}$ from its height over the residue floor $\hat{b}$. The reader admits a verifying mode, in which the tooth is sought within a window around an expected grid, and a blind mode, in which the first tooth beyond the signal lobe is taken. Which mode produced each reported figure is stated with that figure. For mixed grains the empirical characteristic function is evaluated directly on the integer sample, with no binning, at the exclusive replica centers of each candidate grain, the multiples $2\pi m/g$ with $1 \leq m < g$ that are not centers of any finer candidate; the sampling floor $1/n$ is subtracted from the squared modulus; the amplitude of grain g is the root mean of the corrected squared moduli over its exclusive centers; and the weights follow by Möbius differences from the coarsest grain down, each grain's amplitude less the weights of its coarser multiples, with the unit share as one minus the sum. For integer data the unit comb has amplitude exactly one, so the unit class and any unrounded residual are one observational class and are reported as one number. This is the reader of Supplementary Table S8 and of adheaping 1.1.0.

The comb is detected by forming the power spectrum $|\hat{\phi}(w_k)|^2$, removing its mean, and taking its outer-product covariance, a fourth-order statistic. For each candidate replica period K the symmetry defect of the shift-by- K quotient cyclic group is evaluated in closed form from the circular autocorrelation of the spectrum, and the smallest period whose defect falls into a low band, within 0.35 of the way from the minimum to the maximum defect across candidates, is the detected replica period. A spread requirement, that the defects vary across candidates by more than fifteen percent of their maximum, guards against a false detection on exact data, where every period fails equally. The analysis span, the internal grid size, and the candidate period list materially affect whether a given grid is detected. The settings used for Supplementary Table S9 are the library defaults: the span is the data range padded by fifteen percent at each end, the internal grid has 512 cells, and the candidate periods are 8, 16, 24, 32, 48, 64 and 96 in frequency index. With them, over 10 replicates, the detector fires in some replicate in 7 of the sixteen cells and in a majority of replicates in 0, with a highest firing rate of 0.50. In the 5 cells where it fires at $D = 0.25$ or $D = 0.5$ the mean detected grid is within one percent of the imposed one; in the 2 cells where it fires at $D = 1.0$ the detected grid is off

by -71 and -53 percent, the true replica period having failed the defect band while a longer one passed.

The lattice reader's detection statistic λ is the natural logarithm of the ratio of the base-period tooth power to a robust spectral noise floor, a rolling-median lower-quantile floor measured beyond the signal extent so the signal band cannot pollute it. The reader fires only when the tooth amplitude also exceeds four times the floor amplitude. The statistic is a log ratio and not a standardized deviate, so a value of λ should not be read against normal quantiles, and no calibrated threshold on it is claimed.

The mixed-grain reader and the known-truth sweep

A nonnegative least-squares inversion over the whole divisor lattice, fitted to binned replica-center amplitudes, is an alternative design and is not used. Working the characteristic function of each rounding class through, as in the Results, shows why the choice does not lie there. Each class is a lattice variable whose characteristic function is one at every multiple of its own $2\pi/g$, so the amplitude at a center exclusive to g already isolates the classes that share it, and the Möbius differences are the inversion the model specifies rather than one fitted to it. What limits the reading at the cigarette scale is aliasing of the finer classes into those centers when the base is sharp at the coarsest grain, which no choice of numerical design removes, and the sweep of Supplementary Table S8 measures it.

The sweep draws integer counts from a gamma base with standard deviation three quarters of its mean, at means of 12, 24, 48, 96 and 192, assigns each count to a grain with weights 0.18, 0.12 and 0.16 on the five, ten and twenty grains and the remaining 0.54 left at the unit grain, rounds each class to its grain with values that would round to zero placed at the grain, and reads the result with the exclusive-center reader, over 200 simulations per mean at $n = 1019$. At the cigarette scale the base gate on the observed shares of multiples of five, ten and twenty passes. The ratios on the five grain run 1.27, 1.09, 1.03, 1.01, 1.00 and on the twenty grain 0.56, 0.69, 0.89, 0.96, 0.97 as κ runs 0.45, 0.90, 1.81, 3.62, 7.24. A simulation that resamples the already-heaped real counts and rounds them again materially overstates the round-number shares and was not used. On the real counts the exclusive-center amplitudes are 0.124, 0.292 and 0.467 at the exclusive centers of the twenty, ten and five grains.

A model-based refinement is not adopted. Given a base estimate, a nonnegative regression of the empirical characteristic function on the characteristic functions of the base rounded to each grain, over a dense frequency grid, recovers the weights at the cigarette scale when the base is the true one; with the base estimated from the data by drawing each report's grain from its posterior and dequantizing within the cell, iterated, the simulation returns 0.494, 0.217, 0.125 and 0.153 against 0.54, 0.18, 0.12 and 0.16. That is a rounding model and an imputation loop, the lineage this work contrasts itself with, and it is shipped with the reproducibility scripts for comparison only.

Estimating the grid, and reading the heaped fraction

Two experiments measure what the comb supports when the grid is not supplied. Both use the four targets of Table 1 on the binning stated above, at $n = 4000$, and generate their seeds from the rule of record.

The end-to-end experiment (Supplementary Table S10) runs four reconstructions in every cell of Table 1 at 20 replicates. The oracle pipeline is the combined estimator given the imposed D . The verifying and blind pipelines take $\hat{D}$ from the grid reader in its two modes and pass it to the same estimator. The detector pipeline takes $\hat{D}$ from the fourth-order detector in the replicates where it fires. A grid estimate counts as usable when it is finite, larger than two cell widths and smaller than a fifth of the analysis span; an estimate outside that range is recorded as a failure of the pipeline rather than replaced by the true value, so the failure rate is visible in the table. The blind reader returned a usable estimate in 180 of the 320 replicate-cells and the detector in 42; measured against the naive kernel, the do-nothing alternative, the blind pipeline wins in 4 of the sixteen cells where the oracle pipeline wins in 12. The detector fires in some replicate in 8 of the sixteen cells; its single bad firing is Gaussian at $D = 1.00$, where the detected grid is too fine by about seventy percent and the reconstruction costs a factor of 701. The verifying pipeline departs from the oracle by more than three decimals in four cells, the largest departure being 32.1 percent at strongly skewed at $D = 1.50$, where the reader's estimate happens to give the lower error of the two.

The heaped-fraction experiment (Supplementary Table S11) draws a sample from each target, rounds a randomly chosen fraction p of it to the grid D and leaves the rest alone, then reads $\hat{p}$ from the comb with the true D supplied, so the experiment isolates recovery of p . Fractions of 0, 0.2, 0.4, 0.6, 0.8 and 1 are run at grids of 0.5, 1 and 2 over 20 replicates, giving 12 target-grid cells. Each cell is summarized by the slope and the coefficient of determination of $\hat{p}$ regressed on p , by whether the recovery is monotone, and by the value returned at $p = 0$. Fitting one calibration line to all 12 cells together and inverting it leaves errors in p of 0.140 to 0.659 depending on the cell, which is the sense in which a single calibration is not available. Ordering the cells by $D/\sigma_{\min}$, where $\sigma_{\min}$ is the smallest component standard deviation of the target, separates the cells in which the reader behaves as described from those in which it does not.

The NHANES reference bandwidth

The no-heaping reference for the NHANES study is a Gaussian kernel estimate of the unrounded weights at the Silverman bandwidth, 9.681 lb for these 5185 observations on a grid of $M = 4096$ cells. Error against it partly rewards resemblance to that smoothing, so the reference bandwidth was multiplied by 0.5, 0.75, 1, 1.5 and 2 and the study rerun at each (Supplementary

Table S7). The mean rank of the combined estimator among the four methods moves through 1.6, 2.4, 3.4, 3.6, 3.6. The direction is the one the construction predicts, since a band-limited de-heaper preserves fine structure that a more heavily smoothed reference has removed and is penalized for it, and the penalty grows with the reference smoothing. The grid reader, run in its verifying mode with the imposed grid supplied as the expected one, confirms that grid to within 0.6 percent at four of the 5 grids and to 3.0 percent at $D = 40$, where the grid approaches the width of the reference bandwidth.

Figure normalizations, the grid sweep, and a cross-check

The residual plotted in Fig. 1b admits two normalizations and they are not interchangeable. Divided pointwise by $|\phi(w)|$, which is the quantity (3) bounds, the error is 0.0023 at six tenths of the band, 2.02 at nine tenths and 1.000 at the edge. Divided by the supremum of $|\phi|$ over the band it is 0.050 at the edge, smaller by a factor of 19.9, so the panel names the normalization it draws. The pointwise ratio diverges at the zeros of $|\phi|$, where a relative error carries no meaning, so the panel is capped rather than drawn to its maximum. The panel is closed-form replica arithmetic at $D = 0.70$ with the replica sum truncated at $|m| = 8$, so it carries no sampling variation.

The grid sweep of Fig. 2 uses the bimodal target at $n = 4000$ over 50 seeds from the rule of record. Past the identifiability ratio the two components of the combined estimator separate sharply, and the combination follows the wrong one: at those grids the combined estimate is a factor of 7.9 worse than the superposition run alone, so it is the band-capacity cutoff rather than either component that fails there. The margin over the naive estimate falls from 10.9 inside the limit to 1.8 beyond it.

That sweep and the benchmark behind Table 1 share this target, this seed rule and 4 grids while being computed by two separately written scripts, so their overlap checks the pipeline. The 24 values produced by the R library agree to at least 5 significant figures. The SEM column is the exception, moving by as much as 3.5 percent at 2 of the four grids, because `Kernelheaping` draws from a generator the seed rule does not reach; that column is therefore reproducible to about a percent rather than exactly.

Datasets and reproducibility

The synthetic targets of Table 1 are four Gaussian mixtures. Two are taken from the normal-mixture benchmark of Marron and Wand¹², their Gaussian (member 1) and kurtotic unimodal (member 4). The remaining two, a bimodal and a strongly skewed target, are defined in Supplementary Table S1 and are not members of that benchmark. All fifteen members of the benchmark are evaluated in Supplementary Table S2 on the same grid, with the same seeds and the same estimators.

The NHANES 2017–2018 body-measures, demographic, and smoking files are public¹⁷. Adult measured weights are converted to pounds and rounded to imposed grids, and self-reported cigarettes per day are taken from the smoking item. The NHANES analyses are secondary analyses of publicly available, de-identified data and required no ethics approval. The Berlin resident-registration matrix of December 2015 is public open data from the Berlin-Brandenburg statistical office, and age-band counts are pooled over the 447 planning areas.

All experiments derive from the fixed seed of record 20260627. Seeds are generated rather than listed: the seed for replicate s at grid width D is $20260627 + 1000s + \text{round}(100D)$, so the full set of draws behind Table 1 and Supplementary Table S2 is reproduced from that rule. Every value in those tables is emitted from a stored results file rather than typed, and the per-seed integrated squared errors, the paired differences, and a provenance record naming the platform, the R version, the cell count, the cell width, and the elapsed time are shipped with the reproducibility scripts. The generated data and the scripts, including the R library that computes every table and a Python reference implementation of the density estimators, are available at the author's repository, <https://github.com/mitch-thornton/kde-ad-heaping>, and the stored result files behind every v10.3 value, with their provenance records, ship with the reproducibility scripts.

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Data availability

The NHANES 2017–2018 data are publicly available from the U.S. Centers for Disease Control and Prevention, and the Berlin 2015 resident-register data from the Statistical Office of Berlin-Brandenburg. The generated data are available at <https://github.com/mitch-thornton/kde-ad-heaping> and archived at Zenodo under the concept DOI 10.5281/zenodo.21181205, which resolves to the newest deposited version.

Code availability

The reproducibility scripts, including the R library that computes every table and the Python reference implementation, are available at <https://github.com/mitch-thornton/kde-ad-heaping>, with the version used here archived at Zenodo under the concept DOI 10.5281/zenodo.21181205. The R library is also distributed on CRAN as the package `adheaping`; version 1.0.0 is on CRAN and version 1.1.0, which carries the exclusive-center mixed-grain reader, **[PENDING: D-4, state submitted or published once the CRAN 1.1.0 submission is confirmed]**.

Author contributions statement

M.A.T. developed the theory, implemented and ran the experiments, and wrote the manuscript.

Additional information

Competing interests M.A.T. is the inventor on U.S. provisional patent applications, assigned to Clearpoint Research, LLC, relating to the algebraic-diversity methods described in this work.

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Supplementary information accompanies this paper (Supplementary Section S1, Supplementary Fig. S1, Supplementary Tables S1 to S11).